Learning in the Age of AI

Lecture 01: Pedagogic Concept

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Why This Teaching Style

Guiding Question

If AI can already generate summaries, definitions, and polished explanations, what is the job of the lecture?

$$\text{Lecture value} \neq \text{information transfer alone}$$

Why Might This Be True?

- AI compresses and restates efficiently
- students still struggle with meaning, transfer, and judgment
- explanation alone does not guarantee interpretation

Interpretation

The lecture should create tasks that are hard to outsource:

- interpreting formal statements
- critiquing polished answers
- comparing alternative explanations
- exposing hidden assumptions

> The lecture is valuable when it organizes thinking, not when it merely restates information.

The Question-First Module

Guiding Question

Why begin with a difficult-looking result rather than a friendly explanation?

$$\text{Question} \to \text{Tension} \to \text{Interpretation} \to \text{Stabilization}$$

Why Might This Be True?

- questions create attention
- compact results create productive uncertainty
- students must articulate meaning before being told
- synthesis stabilizes what should remain

Interpretation

The first slide in a module should not close the issue. It should open it while already showing the shape of the answer.

> Good module design gives students something to resolve, not only something to receive.

Math First, Then Meaning

Guiding Question

Why show a formal definition, equation, or compact claim before the explanation?

$$\text{formal statement first} \Rightarrow \text{students must interpret}$$

Why Might This Be True?

- it interrupts passive agreement
- it reveals whether students can assign meaning
- it surfaces hidden assumptions

Interpretation

One useful teaching move is: *"This could be an AI answer. What does it actually mean, and what is missing?"*

> Formalism becomes educational when students have to explain it in conceptual language.